

Spiking Neural Dynamics and Plasticity in Small Recurrent Networks for Adaptive Robot Navigation

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Abstract. This study investigates neural dynamics and synaptic plasticity in recurrent spiking networks. Using a single recurrent neuron, we show that modulation of sensory input and recurrent weights shapes activation trajectories, producing temporal memory, hysteresis, and enabling real-time adaptive control. A compact modular network—a two-neuron spiking recurrent core with a non-spiking post-processing layer—was deployed on autonomous robots. Experiments with frontal and corner obstacles demonstrate that environmental geometry drives distinct neural activation, synaptic adaptation, and motion responses. Long-range navigation confirms that excitatory recurrence enhances spike persistence, inhibitory coupling regulates activation, and online plasticity maintains stable, adaptive behavior. These results show that small recurrent networks can efficiently regulate neural dynamics in real time, enabling adaptive navigation with low computational cost for practical neuro-robotics.

Keywords: Neural Dynamics, Adaptive Neural Control, Synaptic Plasticity, Temporal Memory, Embedded Neural Network, Autonomous Robots

1 Introduction

Understanding neural dynamics—how neural states evolve with sensory inputs and parameters such as synaptic weights and network structure—is fundamental for developing adaptive and intelligent systems [2, 4]. It underlies memory, decision-making, and behavior in both biological and artificial systems. Neural dynamics spans multiple levels, from neuron-level responses [11] to network-level memory and decision states [1], and up to embodied behavioral dynamics [6]. It can also be categorized into non-spiking and spiking neural systems (Fig. 1), the latter being more biologically plausible. For non-spiking neural system, several studies have investigated neural dynamics phenomena, properties, functionality, and usability, as well as their relationship to biological neural systems. Neural dynamics have been used to generate delayed activation and signal propagation in multi-segmented robots [10], as well as to prolong control signals for improved obstacle avoidance [5]. These behaviors are closely related to hysteresis or short-term memory, enabling systems to retain and integrate recent inputs and sustain

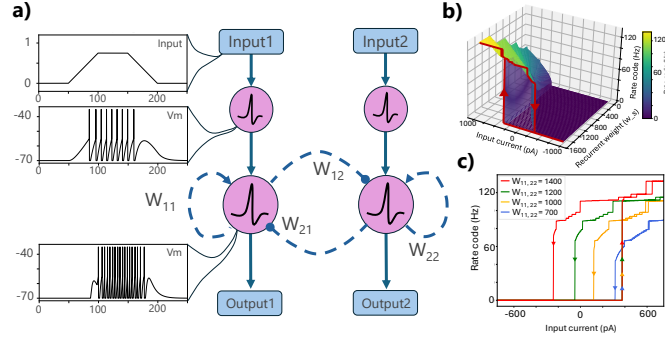


Fig. 1. The graphical abstract shows: (a) a spiking neural control system with signal dynamics at each level of the network, from input to the membrane potential (Vm) of the spiking neuron; (b) a 3D surface heatmap illustrating the hysteresis neural dynamics of a recurrent spiking neuron in the input current–recurrent synaptic weight parameter space; and (c) a hysteresis loop highlighting the hysteresis effect of the recurrent spiking neuron with respect to changes in recurrent synaptic weight.

actions beyond immediate stimuli [9]. Neural dynamics have also been applied to generate oscillatory signals for locomotion via central pattern generators (CPGs) [10, 8], and chaotic signals for adaptive behaviors such as walking pattern [12] and foraging exploration [5]. While non-spiking neural dynamics are well studied at all levels, research on spiking neural dynamics remains comparatively limited.

With the goal of developing low-computational-power neural control systems, we investigate spiking neural dynamics, starting from a single recurrent spiking neuron. Our analysis shows that such neurons exhibit hysteresis and memory-like dynamics, similar to those observed in recurrent non-spiking neurons. Building on this, we develop a spiking neural control network and evaluate how its dynamics influence behavior at both the network and task levels in autonomous robot navigation. Taken together, this study makes the following key contributions:

- Analysis of spiking neural dynamics, highlighting key properties and implications for control design.
- Investigation of how input current, recurrent connectivity, and synaptic parameters shape dynamics such as hysteresis and short-term memory.
- Development of a compact spiking neural control system with synaptic plasticity for adaptive robot navigation.

2 Materials, Methods, and Model Verification

In this section, we describe the system design and the methodology used to analyze, verify, and compare the neural dynamics of both non-spiking and spiking neuron models. In addition, we explain how neural dynamics manipulation is utilized to design and optimize the neural control system, including its learning

mechanism for online adaptation of the recurrent synaptic weight w_s , which is ultimately deployed to adapt robot behavior.

2.1 Neuron models

Neurons are fundamental components of the biological nervous and brain system. They process information by sending electrical signals called action potentials. These signals arise from differences in ion concentrations inside the cell and in the surrounding fluid. Ions move in and out of the neuron through a membrane that can adjust how easily electrical signals pass through it. A biological neuron (see Fig. 2a) consists of four main parts: dendrites, soma, axon, and synapses. Dendrites receive chemical signals and convert them into electrical signals, which are combined in the soma to determine whether an action potential is produced. The axon then carries this signal to other neurons. Communication between neurons occurs at synapses, which can be chemical or electrical. Through synapses, neurotransmitters transmit signals that can either increase (excitatory synapses) or decrease (inhibitory synapses) the activity of the receiving neuron.

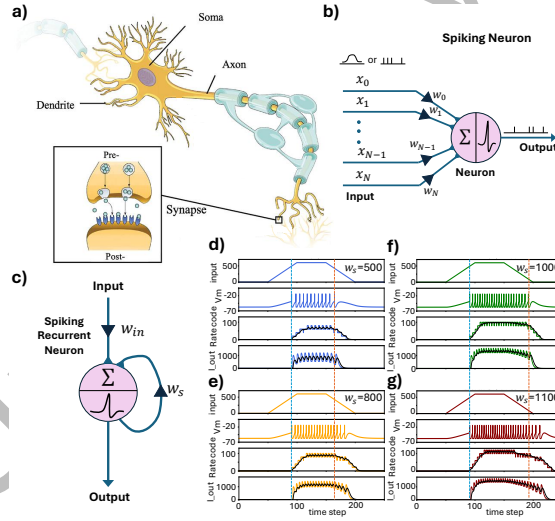


Fig. 2. A comparison of (a) a biological neuron and its components (illustration in (a) reproduced from [7], licensed under CC BY 4.0), (b) a spiking neuron, (c) a recurrent spiking neuron, and (d–g) Input–output responses of the recurrent spiking neuron under recurrent synaptic weights of 500, 800, 1000, and 1100, respectively.

From a computational perspective, biological neurons can be abstractly represented using both artificial (non-spiking) and spiking neuron models, which are widely employed in modern intelligent systems. In this work, we focus on spiking

neuron models and adopt the leaky integrate-and-fire (LIF) neuron model, with parameters based on the NEural Simulation Tool (NEST) [3]. In this model, the membrane potential dynamics, spike generation mechanism, and synaptic current evolution of each neuron are governed by the following equations:

$$\frac{dV_m}{dt} = -\frac{V_m - E_L}{\tau_m} + \frac{I_{\text{syn}} + I_e}{C_m}, \quad (1)$$

where V_m denotes the membrane potential, E_L is the resting membrane potential, which is set to -70 mV, I_{syn} denotes the synaptic input current discussed below, I_e is the sensory input current or bias input, τ_m is the membrane time constant, which is set to 10 ms, and C_m is the capacity of the membrane, which is set to 250 pF.

A spike is generated at time $t^* = t_{k+1}$ when the membrane potential crosses the threshold as below:

$$V_m(t_k) < V_{th} \quad \text{and} \quad V_m(t_{k+1}) \geq V_{th}, \quad (2)$$

where V_{th} is the spike threshold, which is set to -55 mV.

$$I_{\text{syn}}(t) = I_{\text{syn,ex}}(t) + I_{\text{syn,in}}(t), \quad (3)$$

$$I_{\text{syn},X}(t) = \sum_j w_j \sum_k i_{\text{syn},X}(t - t_j^k - d_j), \quad (4)$$

$$i_{\text{syn},X}(t) = \frac{e}{\tau_{\text{syn},X}} e^{-\frac{t}{\tau_{\text{syn},X}}} \Theta(t), \quad (5)$$

$$i_{\text{syn},X}(t = \tau_{\text{syn},X}) = 1, \quad (6)$$

where $I_{\text{syn,ex}}$ and $I_{\text{syn,in}}$ denote the excitatory and inhibitory synaptic input currents, respectively, both governed by Eqs. 3 - 6. The index j refers to presynaptic neurons, which can be either excitatory ($X = \text{ex}$) or inhibitory ($X = \text{in}$), and k indexes the spike time of neuron j . The term d_j represents the synaptic delay from neuron j , and $i_{\text{syn},X}(t)$ denotes the individual post-synaptic currents (PSCs). The $\Theta(x)$ is the Heaviside step function, and the PSCs are normalized to have a unit maximum as defined in Eq. 6. The green-highlighted terms (Eq.s 1 and 4) will be the parameters to investigate the neural dynamics.

2.2 A Single Recurrent Neuron and Neural Dynamics

A single recurrent neuron is a neuron whose output is fed back into its input. This feedback creates intrinsic neural dynamics, enabling the neuron to rely on both past activity and current input. Figures 2c illustrates the computational structures of a single recurrent spiking neuron, in which the membrane potential, spike generation, and synaptic output of the spiking recurrent neuron are governed by Eqs. 1–6. To illustrate the neural dynamics of the recurrent spiking neuron, the synaptic input weight is set to 1.0. The sensory input (I_e in Eq. 1)

is manually defined to start at 0.0, gradually increase to 700.0, remain at this level for a certain period, and then decrease back to 0.0, as shown in Figs. 2d–2g. We then set the recurrent synaptic weight (w_s in Fig. 2c or w_j in Eq. 4, which contributes to I_{syn} in Eq. 1) to 500, 800, 1000, and 1100, respectively, in order to investigate the resulting neural dynamics. The corresponding outputs are presented in Figs. 2d–2g within the relevant time window. The blue, yellow, green, and red curves represent the input and output signals for recurrent synaptic weights of 500, 800, 1000, and 1100, respectively. The vertical blue and orange dashed lines indicate the activation and deactivation times of the output. These markers are used as references to compare the temporal delay in output deactivation with respect to changes in the recurrent synaptic weight, indicating a form of short-term memory behavior.

2.3 Analysis of Recurrent Neural Dynamics

Here, we perform a detailed analysis of recurrent spiking neural dynamics by investigating the output behavior of a recurrent neuron under variations in both sensory input and recurrent synaptic weight. In this work, we focus on excitatory recurrent synapses (positive weights); therefore, inhibitory recurrent synapses (negative weights) are not considered. To explore the neural dynamics of the recurrent spiking neuron, we set the input synaptic weight to 1.0 and perform a sweep⁴ of the sensory input (I_e) from -1000 to 1000 (upward), followed by a reverse sweep from 1000 to -1000 (downward), while varying the recurrent synaptic weight (w_s) from 0 to 1500. By varying these parameters allows us to construct a 3D surface heatmap representing the relationship between sensory input, recurrent synaptic weight, and neuron output. The heatmap encodes the magnitude of the neuron output (firing rate) as a function of both sensory input and recurrent synaptic weight, enabling comparison of activation patterns across the parameter space.

Figures 3a and 3b show front and back views of the 3D surface plots, where the sensory input, recurrent synaptic weight, and firing rate output are represented along the x-, y-, and z-axes, respectively. These plots illustrate how recurrent feedback shapes the neural dynamics landscape. The light-blue and dark-red trajectories indicate activation and deactivation timing during upward and downward sensory input sweeps at recurrent synaptic weights of 0 and 1500, respectively, revealing hysteresis behavior. Figure 3c presents a 2D top-view heatmap summarizing the neuron output across the sensory input–recurrent weight parameter space. For a fixed recurrent synaptic weight of 700, the neuron activates at point A during the upward input sweep and remains active until deactivating at point B during the downward sweep, thereby generating a hysteresis loop. Notably, the spike activation point occurs at the same sensory input value across all recurrent synaptic weights, whereas the output magnitude increases with input strength. In contrast, the deactivation point is progressively

⁴ At each sweep step, the sensory input is increased by 1.0 and held constant for 2000 time steps before moving to the next value, allowing observation of the steady-state response of the recurrent neuron to each input.

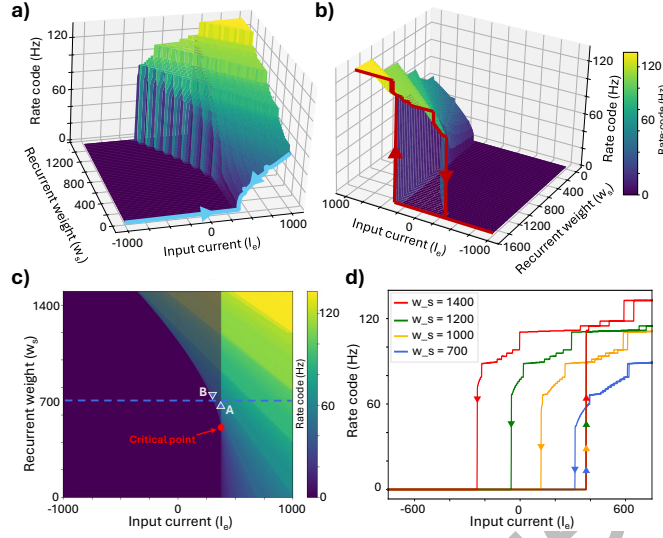


Fig. 3. (a, b) 3D surface heatmaps in the front and back views, showing the response dynamics of the recurrent spiking neuron in the sensory input–recurrent synaptic weight parameter space. (c) A 2D top-view heatmap, illustrating the output response landscape of the recurrent spiking neuron with respect to sensory input and recurrent synaptic weight. (d) A hysteresis loop plot, highlighting the hysteresis effect with respect to the recurrent synaptic weight changes. Note that the critical recurrent weight is 500.

delayed, following the gray shaded region in Fig. 3c. Figure 3d illustrates the corresponding input–output trajectories in terms of firing rate, where increasing recurrent synaptic weight produces progressively larger hysteresis loops. This expansion reflects enhanced output persistence and short-term memory induced by stronger recurrent feedback.

There are two notable characteristics of the hysteresis loops in recurrent spiking neurons that differ from those observed in non-spiking neurons [5]. First, in contrast to the non-spiking case, the recurrent spiking neuron does not exhibit delayed activation as the recurrent synaptic weight varies. Instead, the output is activated at the same input value (i.e., $I_e = 376$ pA). Second, the output magnitude of the recurrent spiking neuron—characterized in terms of spike frequency, firing rate, and synaptic output current—varies with the recurrent synaptic weight. As shown in Figs. 2d–2g and 3, increasing the recurrent synaptic weight leads to higher spike frequency, increased firing rate, and stronger synaptic output current. However, similar to the recurrent non-spiking neuron, the recurrent spiking neuron also exhibits delayed output deactivation. This behavior can be interpreted as persistence or prolongation of the output, whereby the firing state is briefly maintained even after the input no longer drives the neuron.

2.4 Manipulation of Recurrent Neural Dynamics

As discussed in the previous section, the neural dynamics of the recurrent neuron are primarily governed by two key parameters: the (sensory) input and the recurrent synaptic weight, corresponding to the input pathway and the recurrent feedback pathway, respectively. Therefore, in this section, we demonstrate intentional control of recurrent neural dynamics by manipulating these two pathways to regulate the neuron dynamics and temporal response, focusing on the recurrent spiking neuron.

First, we manipulate neural dynamics via the input pathway by adjusting three input characteristics: (1) the sensory input mapping range, (2) the input synaptic weight, and (3) an additive bias current. To interpret the effect of the manipulated mapping range and input synaptic weight, we refer to the diagram in Fig. 3, where the synaptic input axis is rescaled to represent the normalized and computed as the product of the mapped synaptic input and the input synaptic weight. To examine the effect of bias input, we fix the recurrent synaptic weight at 1200 and test bias currents of 0, 500, and -500 pA (Fig. 4a). The zero-bias condition (Fig. 4b, blue color) serves as the baseline. A positive bias of 500 pA (Fig. 4b, purple color) accelerates spike activation, increases firing rate, and delays deactivation, thereby prolonging spiking activity. In contrast, a negative bias of -500 pA (Fig. 4b, brown color) delays activation and accelerates deactivation, reducing spike persistence. From a hysteresis perspective, positive bias shifts the hysteresis loop and region leftward in the sensory input-recurrent weight space, whereas negative bias shifts them rightward. Together, these results show that bias input can systematically regulate activation and deactivation timing of the recurrent spiking neuron.

Second, to manipulate neural dynamics along the recurrent pathway, we introduce synaptic plasticity for the recurrent synaptic weight (w_s), using Hebbian-like learning with a synaptic scaling mechanism as follows:

$$\Delta w_s = \underbrace{\mu \cdot v(t-1) \cdot v(t) \cdot R(t)}_{\text{Hebbian-like learning}} + \underbrace{\gamma \cdot (k - v(t)) \cdot \left(\frac{w_s(t)}{1000}\right)^2}_{\text{Synaptic scaling}}, \quad (7)$$

$$w_s(t+1) = w_s(t) + \Delta w_s, \quad (8)$$

where Δw_s is the recurrent weight plasticity; μ is a learning rate, set to 0.01; γ is a forgetting rate, set to 0.005; k is an offset that enables a constant reduction of the weights when no obstacle is detected, set to -0.01.

$$R(t) = \begin{cases} 1 & I_e(t) > 376, \\ 0 & I_e(t) \leq 376, \end{cases} \quad (9)$$

where $I(t)$ is the reflex signal used as a learning trigger and $I_e(t)$ is the sensory input.

We mapped the firing rate output ($Fr(t)$) to the positive interval $[0, 1]$ by thresholding as below:

$$v(t) = \begin{cases} 1 & Fr(t) > 7.73, \\ 0 & Fr(t) \leq 7.73, \end{cases} \quad (10)$$

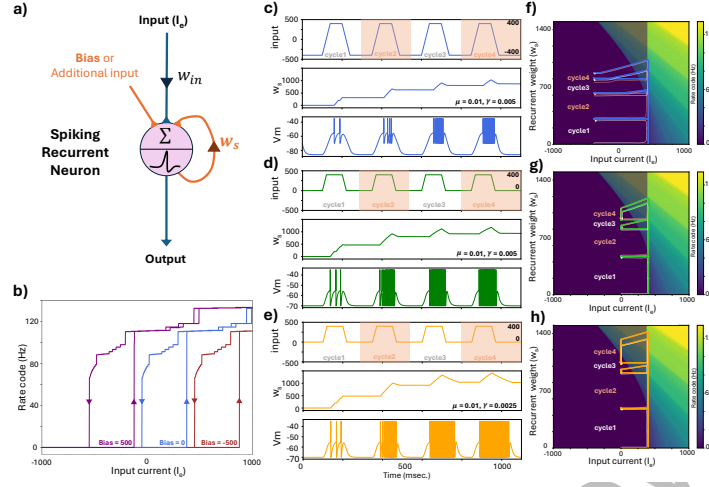


Fig. 4. (a) A recurrent spiking neuron with bias input. (b) Hysteresis loops corresponding to bias currents of 0, +500, and -500, color-coded as blue, purple, and brown, respectively. (c) Input–output response of the membrane potential (V_m) and the corresponding synaptic plasticity of the recurrent synaptic weight (w_s) in the recurrent spiking neuron in Fig. 2c, driven by a real-time input current from -400 to 400 pA. (d) The response and the corresponding synaptic plasticity of recurrent synaptic weight (w_s), driven by a real-time input current from 0 to 400 pA. (e) The response and the corresponding synaptic plasticity of recurrent synaptic weight (w_s), driven by a real-time input current from 0 to 400 pA and change the forgetting rate (γ) to be 0.0025. (f–h) Dynamics trajectory relationship plots of the input current and the synaptic plasticity of recurrent synaptic weight (w_s), corresponding to the result information in (c–e), respectively, overlaying on the hysteresis heatmap in Fig. 3c.

where $v(t)$ is the positive mapped output of the firing rate $Fr(t)$. Note that the firing rate threshold value 7.73 Hz corresponds to the firing rate evoked by the lowest activatable sensory input current value (376 pA) without feedback from the recurrent pathway (i.e., $w_s = 0$).

We apply synaptic plasticity to the recurrent synaptic weight (w_s) in the recurrent spiking neuron (Fig. 4a). To illustrate how synaptic plasticity modulates and shapes the neural dynamics, two sets of parameters involved in the synaptic plasticity mechanism are investigated: (1) the mapping range of the sensory input ($I_e(t)$), which determines the reflex signal ($R(t)$) in the correlation-based Hebbian learning rule, and (2) the learning rate μ and forgetting rate γ . Figures 4c–e show the membrane potential response and the evolution of w_s under real-time input currents repeated over four cycles. Figure 4c uses an input range of -400 to 400 pA with $\mu = 0.01$ and $\gamma = 0.005$, while Fig. 4d uses 0 to 400 pA with the same parameters. Fig. 4e uses 0 to 400 pA with $\mu = 0.01$ and a reduced $\gamma = 0.0025$. The comparison of Figs. 4c and 4d shows that changing the input range from -400 ~ 400 pA to 0 ~ 400 pA drives w_s to higher values, result-

ing in increased spiking activity and prolonged spiking duration. Additionally, the comparison of Figs. 4d and 4e shows that adjusting the learning–forgetting rate ratio—by reducing the forgetting rate from 0.005 to 0.0025—further increases w_s , leading to enhanced spiking activity and longer-lasting responses. Figures 4f–h overlay plasticity-driven w_s trajectories on the neural dynamics heatmap (Fig. 3c) and map them to rate-coded firing responses, showing how input profiles and learning parameters shape synaptic adaptation and neural dynamics. These results demonstrate that the heatmap can guide the design of recurrent learning rules, input currents, and plasticity parameters for compact networks to exhibit temporal delay and short-term memory.

3 Experiments and Results

In this work, we employ a two-neuron recurrent spiking network with synaptic plasticity as the core mechanism for adaptive obstacle avoidance in autonomous robots. By leveraging recurrent neural dynamics, the system enables adaptive behavior control and is evaluated on simulated wheeled and flying robots.

3.1 Experimental Setup

To evaluate the spiking neural control system, we propose a compact obstacle-avoidance network (Fig. 5a) with five layers—sensory input, spiking computation, spike counting, thresholding, and premotor neurons—tested in CoppeliaSim on wheeled and flying robots. The spiking computation layer (Fig. 5b) processes two frontal inputs (Input1, Input2), mapped from 50–0 cm to $[0, 2]$ and encoded by I1 and I2. The core is a two-neuron recurrent network (S1, S2) with plastic excitatory (W_{11} , W_{22}) and inhibitory (W_{12} , W_{21}) synapses. Spikes are counted every 20 ms (50 Hz) and thresholded. A parallel lateral pathway (without recurrent feedback, $w_s = 0$) follows the same pipeline. Frontal and lateral outputs are fused by premotor neurons to generate roll, pitch, and yaw commands. Figures 5c–f illustrate test scenarios with frontal obstacles, corner walls, and distributed environments

3.2 System Performance and Adaptive Behaviors

Robot performance was evaluated across multiple scenarios. Figure 6a shows the wheeled robot’s trajectory during frontal obstacle avoidance, while Fig. 6b presents the corresponding time-series signals, including sensory inputs, spike counts, yaw commands, and synaptic weights. Upon sensory input, the recurrent network generates spikes that are counted, thresholded, and converted into yaw commands, while synaptic weights adapt online. During the initial phase (blue region), similar inputs and low weights produce weak spiking and unstable yaw oscillations, resulting in minimal orientation change. As the robot slightly turns, input asymmetry increases, amplifying spike activity and inducing a stronger corrective turn, enabling successful obstacle avoidance (orange region).

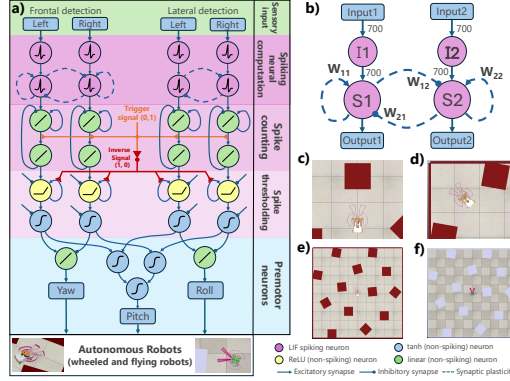


Fig. 5. (a) A compact neural obstacle avoidance network. (b) A spiking neural computation module. (c) A frontal obstacle. (d) A corner-shaped wall. (e) A global environment map with distributed obstacles for evaluating the adaptive wheeled robot. (f) A global environment map with distributed obstacles for evaluating the adaptive flying robot.

Figure 6c shows the robot trajectory for corner-obstacle avoidance, and Fig. 6d presents the corresponding signals, including sensory inputs, spike counts, yaw commands, and synaptic weight adaptation. While the processing pipeline is similar to the frontal case, the behavior differs due to obstacle geometry. During the initial phase (blue region), low synaptic weights lead to weak spiking and unstable orientation. As learning progresses, weights increase, producing alternating left–right turns (green region). Once sufficient, the robot executes a large turn and escapes the corner. This behavior arises from two mechanisms: excitatory recurrent weights (W_{11} , W_{22}) prolong spike activity, while inhibitory weights (W_{12} , W_{21}) suppress competing responses. Their interaction enhances spike contrast and generates decisive yaw commands (orange region). After exiting the corner, synaptic weights decay toward nominal values, preventing overlearning. Figures 6e–h show 30-minute navigation and synaptic plasticity profile for mobile and flying robots in a $10 \times 10 \text{ m}^2$ environment with distributed obstacles.

4 Conclusion

In this work, we investigate the neural dynamics of spiking neuron (LIF model), starting from a single recurrent neuron, we show how modulation of sensory input and recurrent weights shapes temporal memory and hysteresis, and how these dynamics can be adapted online via manipulation of input and synaptic plasticity. A compact architecture—a two-neuron spiking recurrent network with a non-spiking post-processing layer—was implemented and evaluated on mobile robots. Experiments demonstrate that different obstacle geometries induce distinct neural activation, synaptic adaptation, and navigation behaviors.

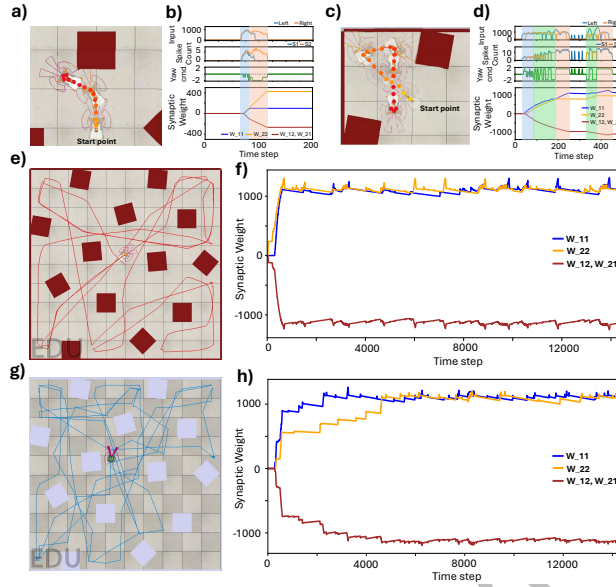


Fig. 6. (a) Movement trajectory of the wheeled robot during frontal-obstacle avoidance, with (b) corresponding sensory inputs, spike-counted outputs, yaw control commands, and synaptic weights. (c) Movement trajectory during corner-shaped obstacle avoidance, with (d) corresponding sensory inputs, spike-counted outputs, yaw control commands, and synaptic weights. (e) Wheeled-robot trajectory in a long-range navigation task inside a $10 \times 10 \text{ m}^2$ enclosed environment with distributed obstacles, and (f) associated synaptic weights. (g) Flying-robot trajectory in the long-range navigation task inside a $10 \times 10 \text{ m}^2$ enclosed environment with distributed obstacles, and (h) associated synaptic weights.

Long-range navigation in a $10 \times 10 \text{ m}^2$ environment confirms that excitatory recurrence enhances spike persistence, inhibitory connections regulate activation, and plasticity prevents overfitting while maintaining stability. These results show that a minimal recurrent architecture enables efficient online learning with low computational cost, suitable for real-time neurorobotics. Future work will deploy this system on microcontrollers or neuromorphic hardware and extend it with advanced perception for real-world mobile and flying robots.

Acknowledgment

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